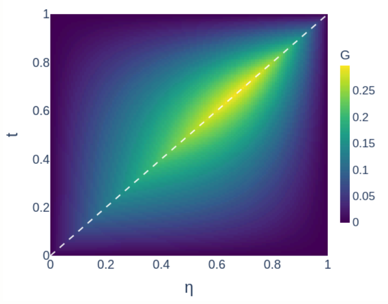
Numerical Evidence I: GreenNet Kernel Approximation

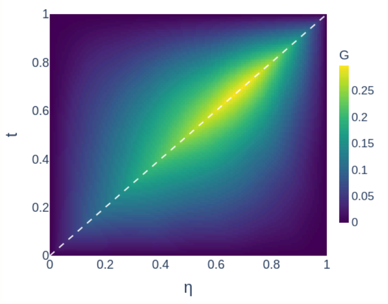
Capturing the singular Green structure on an axial interval

$$\Omega = \{x^2 + y^2 < 0.5^2\}, \qquad -\nabla \cdot (a\nabla u) = f, \qquad a(x,y) = 1 + \frac{1}{2}\sin(2\pi x)\sin(4\pi y), \qquad \mathbf{b} = 0,\ c = 0.$$

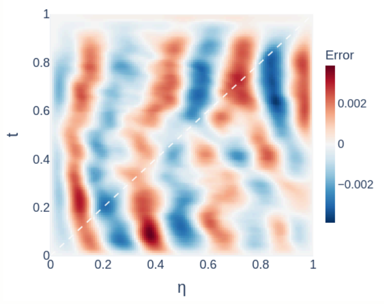
REFERENCE KERNEL



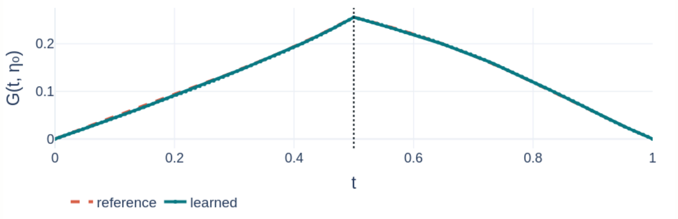
LEARNED KERNEL



SIGNED ERROR



FIXED-H SLICE



DIAGNOSTICS

η	0.50
Kernel rel. L2	1.11%
Slice rel. L2	0.92%
Diagonal band	|t-η| ≤ 5/128 = 0.0391
Error mass / band area	9.5% / 8.3%
Mean error / off-band mean	1.16x

The learned kernel captures the singular Green structure, and the signed error is not concentrated near the singular diagonal.